

Positive Square Energy of Hexacyclic Graphs

Conditional manuscript skeleton—draft, not a theorem announcement

AI-generated draft; no human authorship is claimed

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Draft status. The headline implication in Theorem 1.1 is explicitly conditional. This manuscript does not claim that every hexacyclic graph has the stated property. Its finite premise is not discharged here. In particular, promotion is blocked until the order-nine and order-ten exact owners, the rank-six orders-2–10 master, and the all-connected synthesis master all pass their specified full normal and optimized replay gates.

Abstract

Let G be a finite simple connected graph with n vertices and $n + 5$ edges. This draft records an analytic reduction of the desired inequality $s^+(G) \geq n$ to two explicitly stated proof premises. A repository ledger is cited for the multiblock premise, but this manuscript does not reproduce or independently validate that ledger. In the unique positive-rank cyclic-block case, suppression gives one of 1198 rank-six multigraph kernels, with order distribution 1, 4, 26, 84, 216, 314, 325, 162, 66 on two through ten branch vertices. Exact path elimination reduces arbitrary subdivisions to physical parity orbits and canonical-plus-coordinate frontiers, and one-vertex additivity includes arbitrary rooted trees. The final single-block premise is not yet available: in particular, incomplete order-nine or order-ten coverage is not a theorem owner, and the orders-2–10 and all-connected masters are not green. Reproducibility appendices below therefore contain a mixture of commands for present repository artifacts and clearly marked promotion placeholders, not a publication record or a certificate of the headline conclusion.

1 Scope and conditional statement

All graphs in the spectral statements are finite, simple, and undirected. For the adjacency eigenvalues $\lambda_1, \dots, \lambda_n$ of G , put

$$s^+(G) = \sum_{\lambda_i > 0} \lambda_i^2, \quad s^-(G) = \sum_{\lambda_i < 0} \lambda_i^2, \quad \beta(G) = |E(G)| - |V(G)| + 1.$$

A connected graph with $|E(G)| = |V(G)| + 5$ has cyclomatic rank six.

Positive and negative square energies were introduced in connection with spectral coloring bounds by Elphick, Farber, Goldberg, and Wocjan [1]. Liu, Tang, and Zhang proved the conjectured general bound $\min\{s^+(G), s^-(G)\} \geq n - 1$ for connected n -vertex graphs [3]. The conclusion considered here is instead the stronger edge-surplus conjecture of Akbari, Kumar, Mohar, Pragada, and Zhang: a connected n -vertex graph with at least $n + 1$ edges should satisfy $s^+(G) \geq n$ [2, Conjecture 1.2]. The present draft neither proves that conjecture in general nor unconditionally proves its hexacyclic case.

Theorem 1.1 (Conditional main implication—draft only). *Assume both of the following premises.*

1. *The mathematical assertions in the cited owner-exact multiblock ledger are correct and cover every finite simple connected rank-six graph having at least two positive-rank cyclic blocks.*
2. *For every one of the 1198 suppressed rank-six kernels, every physical parity orbit and every required canonical-plus-coordinate target has an exact proof owner establishing the budget and certificate semantics described in Proposition 6.1; the order-nine and order-ten owners and the orders-2–10 implication master pass full exact replay in both normal and optimized Python modes; and an all-connected synthesis master checks the two scopes as disjoint and exhaustive. Regenerated target keys must equal a disjoint union of final owner-key sets; labels, counts, hashes, and successful process exits alone do not satisfy this premise.*

Then every finite simple connected graph G with $|V(G)| = n$ and $|E(G)| = n + 5$ satisfies

$$s^+(G) \geq n.$$

Remark 1.2 (Current logical status). Theorem 1.1 is an implication, not an unconditional theorem. At the time of this draft the required order-nine/order-ten promotion owners and master acceptance records are not all present. A coverage count, search-completion field, process exit status, or digest without exact replay and exact owner-set equality does not discharge the premise.

No proof of either premise, strict inequality, equality classification, edge-addition monotonicity, one-edge-subdivision monotonicity, result for nonsimple realizations, human review, independent verification, peer review, publication, or external dissemination is claimed.

2 Analytic foundations

For a finite graph $H = (V, E)$ define

$$\kappa(H) = \min_{\substack{R \succeq 0 \\ R_{vv}=1 \ (v \in V)}} \sum_{uv \in E} \frac{2}{1 - R_{uv}}, \quad (1)$$

where a summand is $+\infty$ if $R_{uv} = 1$.

Lemma 2.1 (DNN witness implication). *For every finite simple graph H ,*

$$s^-(H) \leq \kappa(H).$$

Consequently, any feasible correlation matrix with objective at most b proves $s^+(H) \geq 2|E(H)| - b$.

Proof. Write the adjacency matrix as $A = P - N$, where $P, N \succeq 0$ are its positive and negative spectral parts, and put $M = N \circ N$. The Schur product theorem gives $M \succeq 0$ and $M \geq 0$ entrywise. Moreover,

$$\langle J, M \rangle = \text{tr } N^2 = s^-(H), \quad \frac{s^-(H)}{2} = - \sum_{uv \in E(H)} N_{uv} \leq \sum_{uv \in E(H)} \sqrt{M_{uv}}.$$

For every feasible R ,

$$\langle J, M \rangle - 2 \sum_{uv \in E(H)} (1 - R_{uv}) M_{uv} = \langle R, M \rangle + 2 \sum_{\substack{u < v \\ uv \notin E(H)}} (1 - R_{uv}) M_{uv} \geq 0.$$

Weighted Cauchy–Schwarz and cancellation when $s^-(H) > 0$ give

$$s^-(H) \leq \sum_{uv \in E(H)} \frac{2}{1 - R_{uv}}.$$

The zero and infinite-objective cases are immediate. Minimize over R and use $s^+(H) + s^-(H) = \text{tr } A^2 = 2|E(H)|$. $\square$

Lemma 2.2 (Exact path cost). *If a kernel edge with endpoint correlation r is replaced by a path of length $q \geq 1$, its least DNN contribution in excess of its q edges is*

$$f_q(r) = q \tan^2 \left(\frac{\arccos((-1)^q r)}{2q} \right).$$

For fixed r and parity, $f_{q+2}(r) \leq f_q(r)$, with endpoint values understood in the extended sense.

Proof. Represent the endpoints by unit vectors and alternately negate vectors along the path. The transformed endpoint angle is $\gamma = \arccos((-1)^q r)$. The spherical triangle inequality and convexity of $\sec^2(x/2)$ make q equal angles optimal, giving $q \sec^2(\gamma/(2q)) = q + f_q(r)$. With $z = \gamma/(2q)$,

$$\frac{d}{dq} \left[q \tan^2 \left(\frac{\gamma}{2q} \right) \right] = \tan z \sec^2 z (\sin z \cos z - 2z) \leq 0.$$

One-sided limits handle endpoint cases. $\square$

Lemma 2.3 (One-vertex additivity). *If $H = H_1 \vee H_2$ is a one-vertex sum, then $\kappa(H) = \kappa(H_1) + \kappa(H_2)$. In particular, $\kappa(T) = |E(T)|$ for every finite tree T .*

Proof. Restriction gives one inequality. For the other, identify the common unit vector in minimizing Gram representations and place the two perpendicular components in mutually orthogonal subspaces. The objectives add. Every tree edge costs at least one, and opposite unit vectors on the two bipartition classes attain one on each edge. $\square$

3 Block decomposition

We use the standard block convention: the blocks of a connected graph are its maximal 2-connected subgraphs together with its bridge copies of K_2 . A *positive-rank cyclic block* is a block B with $\beta(B) > 0$. Thus “a unique positive-rank cyclic block” does not mean “one nontrivial block”: attached trees contribute nontrivial K_2 blocks.

Lemma 3.1 (Cyclic-block rank additivity). *The cyclomatic rank of a connected graph is the sum of the cyclomatic ranks of its positive-rank cyclic blocks.*

Proof. Following the block–cut tree, adjoining a block B at one cut vertex adds $|V(B)| - 1$ vertices and $|E(B)|$ edges, and hence adds $|E(B)| - |V(B)| + 1 = \beta(B)$ to the rank. Bridge blocks contribute zero. $\square$

Thus the positive cyclic-block ranks of a rank-six graph form exactly one of

$$\begin{aligned} 1^6, \quad 2 + 1^4, \quad 2 + 2 + 1 + 1, \quad 2 + 2 + 2, \quad 3 + 1 + 1 + 1, \quad 3 + 2 + 1, \quad 3 + 3, \\ 4 + 1 + 1, \quad 4 + 2, \quad 5 + 1, \quad 6. \end{aligned} \tag{2}$$

The first ten alternatives are the multiblock branch; the final alternative has one positive-rank cyclic block. This is a structural dichotomy only and does not by itself prove either branch.

Lemma 3.2 (Form of the unique-block branch). *If a connected graph has a unique positive-rank cyclic block B , then every other block is a bridge. The graph is obtained from B by one-vertex-summing finite rooted trees at vertices of B .*

Proof. Every other block has rank zero. Under the stated block convention, a rank-zero block of a simple graph is a bridge K_2 . Delete the node for B from the block-cut tree. Each remaining component is a tree of bridge blocks attached through exactly one cut vertex of B ; two attachments to B would create a cycle in the block-cut tree. Expanding its bridge blocks gives the asserted rooted tree. After suppression is reversed, an attachment vertex may be an internal vertex of a replacement path. $\square$

4 Conditional multiblock interface

Proposition 4.1 (Multiblock branch under Premise 1). *Assume Premise 1 of Theorem 1.1. Every finite simple connected rank-six graph with at least two positive-rank cyclic blocks satisfies $s^+(G) \geq |V(G)|$.*

This proposition is an invocation of the premise, not a proof of the imported ledger. The cited multiblock audit [5] states the asserted scope and supplies the repository argument on which Premise 1 is based. Its scope consists of the first ten rows of (3). It separates direct DNN rows, structural pre-sieve rows, and the packet residuals A–I. Shared cuts, repeated and nested owners, positive bridge connectors, and arbitrary rooted branches are assigned before credit is charged. The direct rank-five-equality-plus-triangle boundary is nonstrict and is not promoted to a strict claim. This skeleton cites that internal exposition rather than duplicating its long owner-by-owner argument. A successful ledger replay checks the encoded predicates; it does not by itself establish that every encoded structural lemma is mathematically sound.

5 Unique positive-rank block

Lemma 5.1 (Rank-six suppression). *Let B be a finite simple 2-connected graph with $\beta(B) = 6$. Suppressing every degree-two vertex produces a loopless 2-connected multigraph K of minimum degree at least three such that*

$$|E(K)| = |V(K)| + 5, \quad 2 \leq |V(K)| \leq 10.$$

Conversely, B is a simple realization obtained by replacing kernel edges by positive-length, internally disjoint paths meeting only at prescribed branch endpoints.

Proof. Suppression preserves $|E| - |V|$, connectedness, and rank. A loop would come from a cycle meeting the remainder only at one branch vertex, contradicting 2-connectivity. If $k = |V(K)|$, minimum degree three and $|E(K)| = k + 5$ give $3k \leq 2k + 10$, so $k \leq 10$; loopless 2-connectivity gives $k \geq 2$. Reversing suppression gives the stated path model. Simplicity is a separate required condition, especially in parallel classes. $\square$

Proposition 5.2 (Exact kernel census). *The rank-six suppressed kernels have the following repository-checked distribution.*

$ V(K) $	2	3	4	5	6	7	8	9	10
$\#K$	1	4	26	84	216	314	325	162	66

There are 1198 isomorphism classes in total.

The exact integer census regenerates multiplicity vectors, rejects supports with a cut vertex, canonicalizes under isomorphism, and compares the complete sorted output with `research/fixtures/rank-six-kernels.json`. No floating-point or spectral calculation enters this census. Its reproducible command is listed in Appendix A.

6 Conditional finite interface

Here an *exact DNN certificate of excess at most five* means an explicit finite record which re-constructs a positive-semidefinite Gram matrix of unit vectors on the realized core, verifies every shared-endpoint identity and every edge denominator $1 - R_{uv} > 0$, and proves by exact integer, rational, or symbolically justified arithmetic that

$$\sum_{uv \in E} \frac{2}{1 - R_{uv}} \leq |E| + 5.$$

A structural owner is admissible only if it has a separately stated proof of the same final spectral conclusion for its exact scope; it is not a DNN certificate merely because a verifier assigns it a target key. Numerical optimization output, a symbolic tag without checked identities, or a stored decimal approximation is not an exact certificate.

For each parallel class of multiplicity m with o odd replacement paths, the canonical simple lengths are

$$c(m, o) = \begin{cases} (2, \dots, 2), & o = 0, \\ (1, \underbrace{3, \dots, 3}_{o-1}, \underbrace{2, \dots, 2}_{m-o}), & o > 0. \end{cases}$$

Simplicity permits at most one unit path in a parallel class. After permuting parallel physical edges, every realization vector ℓ in the parity row satisfies $c_i \leq \ell_i$ and $\ell_i - c_i \in 2\mathbb{Z}_{\geq 0}$.

Proposition 6.1 (Conditional exact-owner interface). *Assume every physical parity orbit of every kernel in Proposition 5.2 is owned either by an exact coarse DNN certificate at its canonical vector whose fixed branch Gram remains within excess five after exact path elimination under every same-parity coordinate lengthening, or by exact DNN certificates (or explicitly typed structural owners proving the same conclusion) for every target*

$$\mathcal{F}(c) = \{c\} \cup \{c + 2\mathbf{e}_i : 1 \leq i \leq |E(K)|\}.$$

For a structural owner, assume additionally an exact all-length and rooted-tree lift covering the descendants for which it claims ownership; Lemma 2.2 may be used only for a DNN owner with a retained branch Gram. Assume that regenerated target keys equal the disjoint union of checked final-owner keys, with no missing, duplicate, or extra keys. Then every simple subdivision of every rank-six kernel, with arbitrary finite rooted trees attached at branch or internal subdivision vertices, satisfies $s^+(G) \geq |V(G)|$.

Proof. If $\ell = c$, use the canonical target. Otherwise choose i with $\ell_i \geq c_i + 2$ and start from the owner of $c + 2\mathbf{e}_i$. For a DNN owner, retain its branch vectors, replace the internal vectors independently on every lengthened path, and apply Lemma 2.2; the excess cannot increase. A typed structural owner instead uses the all-length and rooted-tree implication required in the premise. Thus in the DNN case the subdivided core B , with L edges, has $\kappa(B) \leq L + 5$.

In that DNN case, if attached rooted trees contain t edges, Lemma 2.3 gives $\kappa(G) \leq L + 5 + t$. Rank six gives $|V(G)| = L - 5 + t$, and Lemma 2.1 yields

$$s^+(G) \geq 2(L + t) - (L + 5 + t) = L - 5 + t = |V(G)|.$$

In the structural case, the all-length and rooted-tree conclusion stipulated in the premise gives the same inequality directly. Different frontier targets may use different Gram matrices. No parity-changing subdivision or spectral monotonicity is used. $\square$

The premise of Proposition 6.1 is the open promotion gate in this draft. Orders two through eight having theorem material does not authorize silently extending the statement through orders nine and ten. In particular, current coverage verifiers are not substitutes for the missing promotion owners or the fail-closed orders-2–10 master.

7 Conditional assembly

Proof of Theorem 1.1. By Lemma 3.1, the positive block ranks have exactly the eleven forms in (3). If there are at least two positive-rank cyclic blocks, apply Proposition 4.1. Otherwise there is one rank-six cyclic block. Lemma 3.2, Lemma 5.1, Proposition 5.2, and the assumed exact-owner premise allow Proposition 6.1. The two cases are disjoint and exhaustive. This proves the implication, while making no assertion that its currently blocked finite premise has been discharged. $\square$

A Reproducibility ledger and placeholders

A.1 Commands for present repository artifacts

From the repository root, the kernel census and multiblock ledger can be replayed in normal and optimized modes:

```
python3 research/rank-six-kernel-census-verifier.py
python3 -O research/rank-six-kernel-census-verifier.py
python3 research/hexacyclic-multiblock-ledger-verifier.py
python3 -O research/hexacyclic-multiblock-ledger-verifier.py
```

The lower single-block owner currently exposed by the repository may likewise be audited with

```
python3 research/rank-six-order2-7-master-verifier.py
python3 -O research/rank-six-order2-7-master-verifier.py
python3 research/rank-six-order-eight-kernel-theorem-verifier.py --full
python3 -O research/rank-six-order-eight-kernel-theorem-verifier.py --full
```

A.2 Blocked promotion checks

The following command is deliberately fail-closed until all required promotion owners and frozen outputs exist:

```
python3 research/rank-six-order2-10-master-verifier.py
python3 -O research/rank-six-order2-10-master-verifier.py
```

Current nonzero exit is a blocker, not a successful theorem test. The order-nine and order-ten coverage commands may report partial exact progress, but their output must not be cited as a theorem owner:

```
python3 research/rank-six-order-nine-coverage-verifier.py
python3 -O research/rank-six-order-nine-coverage-verifier.py
python3 research/rank-six-order-ten-coverage-verifier.py
python3 -O research/rank-six-order-ten-coverage-verifier.py
```

Promotion placeholders (intentionally unresolved). Before the conditional label can be removed, replace each item below with a committed, independently regenerated record checked against the manuscript.

1. **TODO-ORDER9-OWNER:** theorem-owner path, schema, source SHA-256, canonical full-replay output SHA-256, exact owner counts, and normal/optimized byte-identity record.
2. **TODO-ORDER10-OWNER:** theorem-owner path, schema, source SHA-256, canonical full-replay output SHA-256, exact owner counts, and normal/optimized byte-identity record.
3. **TODO-RANK6-MASTER:** green orders-2–10 implication manifest, all four child-owner identities, census identity, transitive root, and hostile mutation report.
4. **TODO-ALL-CONNECTED-MASTER:** green two-branch synthesis manifest, scope-disjointness audit, exact conclusion, transitive root, and hostile mutation report.
5. **TODO-CLEAN-REPLAY:** interpreter version, platform, commands, wall-clock/storage notes, complete logs, and normal versus optimized output digests from a clean artifact directory.

No placeholder is evidence, and no digest by itself proves a mathematical predicate.

A.3 Build

This draft is compiled from the repository root with

```
bash scripts/build-paper.sh all-hexacyclic-graphs
```

Build success checks only the LaTeX artifact. It does not turn a red finite gate green and does not promote Theorem 1.1.

B Computational certificate specification

This appendix specifies the formats and the acceptance algorithm intended for the order-nine and order-ten finite premise. It documents the current experimental streams; it does not assert that their manifests cover the full residual ranges, that all decoded targets are owned, or that a promotion master has accepted them. In particular, an **unresolved** record is a valid encoding but is never an accepting proof record.

B.1 Common conventions and target keys

All integers below are mathematical integers, not machine words. A *uvarint* writes a nonnegative integer in base 128, least significant group first, with bit 7 set exactly when another group follows. A signed integer z is first mapped to $2z$ for $z \geq 0$ and to $-2z-1$ for $z < 0$, then written as a uvarint. Multi-byte integers are therefore independent of host endianness. At promotion the decoder must additionally reject a uvarint with a redundant final zero group; the current R10G1 decoder enforces this by decode-re-encode byte equality, whereas the current R9G1 auditor does not yet impose that canonicity check. This difference is a promotion blocker, not a license to accept two encodings of one certificate.

The frozen kernel fixture, in its listed order, regenerates the residual source stream. A source index j denotes row j of that stream (zero based), not a kernel identifier stored in a pack. The verifier reconstructs from j the kernel, its ordered physical paths, the parity row, canonical lengths, and the $p+1$ frontier labels

$$C, X_0, \dots, X_{p-1}, \quad p = 14 \text{ for order nine, } p = 15 \text{ for order ten.}$$

Thus an exact target key is the pair (j, C) or (j, X_i) . A pack owns a half-open source interval $[a, a+r)$, encoded by uvarints a, r . Keys are never accepted from asserted counts: the master regenerates them and compares sets.

Every stored stereographic parameter vector has dimension $d-1$, where $d \in \{9, 10\}$. One positive common denominator D precedes a witness and each coordinate is a signed-varint numerator n , denoting n/D . Vector and path counts are implicit in the regenerated source row; no untrusted dimension field controls allocation or verification.

B.2 The R9G1 wire format

After XZ decompression, an order-nine pack is the following byte concatenation:

```
"R9G1" | fixture_sha256[32] | uvarint(start) | uvarint(count)
      | record[0] | ... | record[count-1].
```

The 32-byte digest is binary, not 64 hexadecimal characters. Here $d = 9$, $p = 14$, and a target bitmap has 15 significant bits. Record tags are

byte	name	body and meaning
0	unresolved	empty; owns no target
1	shared	one shared rational witness, described below
2	template	empty; invokes the exact K971 signed-five-cycle audit
3	individual	uvarint bitmap, then one witness per set bit

A shared body is D , nine branch parameter vectors, then, in physical-path order, all canonical internal vectors and all length-plus-two internal vectors. For a canonical path of length q_i , these two lists have respectively $q_i - 1$ and $q_i + 1$ vectors. The same branch vectors are used for all 15 frontiers. An individual body begins with a nonzero bitmap smaller than 2^{15} ; for each set bit in increasing target order it stores a fresh D , nine branch vectors, and $q_i - 1$ internal vectors on each path, except that the selected X_i path has $q_i + 1$. The template tag is legal only on the regenerated K971 template row. The decoder rejects bad magic or fixture hash, unknown tags, zero denominators, out-of-range intervals or bitmaps, truncation, template/tag mismatch, and trailing raw bytes.

B.3 The R10G1 wire format

The order-ten header is identical with magic **R10G1**; here $d = 10$, $p = 15$, and the bitmap has 16 significant bits. Tags 0–3 have the same layout, with tag 3 named **fallback**. The current version also reserves payload-free, source-row-dependent exact recognizers:

byte	name	required independent audit
4	structural	the typed structural Gram sieve
5	atom	the frozen atom-ledger classification and exact atom audit
6	balanced	the balanced rank-one recognizer

Tag 2 invokes only the regenerated K1133 signed-five-cycle row. Empty tags do not mean “trust the classification”: their recognizers rebuild the required combinatorics and exact Gram or structural predicate. The **R10G1** decoder rejects the malformed cases listed for **R9G1** and requires the raw input to equal its unique re-encoding. XZ is a transport envelope only; acceptance is defined on the decompressed bytes and the exact records they denote.

B.4 Hardware-independent exact arithmetic

For a parameter row $t = (t_1, \dots, t_{d-1}) \in \mathbb{Q}^{d-1}$ the verifier constructs, without floating point,

$$u(t) = \frac{(1 - \langle t, t \rangle, 2t_1, \dots, 2t_{d-1})}{1 + \langle t, t \rangle}.$$

Python arbitrary-precision integers and normalized **Fraction** values implement the field operations. For consecutive unit vectors x, y in the alternating-sign path chain it requires $1 + \langle x, y \rangle > 0$ and adds

$$\frac{1 - \langle x, y \rangle}{1 + \langle x, y \rangle}.$$

The final comparison is the exact cross-multiplied rational inequality $\sum \text{cost} \leq 5$. Endpoint vectors are reconstructed from the common branch list, with the terminal branch negated exactly when prescribed by path parity. Consequently unit norm, shared endpoints, path widths, denominator signs, and the budget are rational identities or integer comparisons. Search seeds, floating-point optimizer scores, wall-clock times, CPU vector instructions, byte order, and native integer overflow cannot affect acceptance. A conforming alternative verifier may use any big-integer rational library, but must produce the same accepted key set from the same raw bytes. XZ decompression, SHA-256, and Python itself remain implementation dependencies; this arithmetic claim is not a claim of independent software verification.

B.5 Fail-closed verifier algorithm

For each order, the promotion verifier shall execute the following algorithm. Every failure aborts with nonzero status; no **assert**, status string, or stored total is an acceptance condition.

1. Hash and parse the pinned kernel fixture, verifier sources, recognizer sources, census output, and canonical manifest; reject any unpinned byte sequence or noncanonical manifest JSON.
2. Regenerate the complete ordered residual source stream and, from it, the complete expected target-key set. Compare its count and ordered SHA-256 with the pinned values.

3. Read manifest chunks in increasing half-open ranges. Require paths to remain below the manifest directory, and ranges to be nonempty, contiguous, nonoverlapping, and exactly $[0, N)$. Reject repeated paths.
4. For each chunk, check compressed byte count and SHA-256, decompress one XZ stream, check raw byte count and SHA-256, parse the matching embedded interval, and reject malformed or noncanonical records and trailing bytes.
5. For every rational record, reconstruct every relevant frontier chain and perform the exact arithmetic audit above. For every payload-free record, run its named exact recognizer. Map only successfully checked targets to their typed owner; map no target from an unresolved bit or tag.
6. Regenerate the separately pinned symbolic-owner keys. Check that each is proved by its typed recognizer, that rational and final symbolic ownership are disjoint after the prescribed precedence rule, and that their union equals the expected target set exactly. Explicitly reject missing, duplicate, and extra keys.
7. Emit canonical ASCII JSON containing scopes, dependency digests, mode and owner counts, ordered key digests, exact partition equality, and the conditional finite conclusion. Replay under `python3` and `python3 -O`; require zero exits and byte-identical reports. The orders-2–10 master must then validate, rather than copy, these two order conclusions.

Steps 1–4 alone are a digest audit, not certificate verification. Likewise, successful verification of one chunk proves only the records in that chunk; aggregating receipts does not replace a fresh full-manifest exact replay.

B.6 Segmented reproducibility

Large packs may be replayed independently by manifest chunk. From the repository root, first perform the inexpensive range and byte audit:

```
python3 positive-square-energy/experiments/rank6_order9_pack_auditor.py \
  --manifest positive-square-energy/experiments/rank6_order9_search_manifest.json \
  --digest-only
python3 positive-square-energy/experiments/rank6_order10_pack_auditor.py \
  --manifest positive-square-energy/experiments/rank6_order10_search_manifest.json \
  --digest-only
```

For each manifest chunk index $i = 0, 1, \dots, m - 1$, use a new output path:

```
python3 positive-square-energy/experiments/rank6_order9_pack_auditor.py \
  --manifest positive-square-energy/experiments/rank6_order9_search_manifest.json \
  --chunk-index i --write-chunk-receipt /tmp/r9-replay/chunk-i.json
python3 positive-square-energy/experiments/rank6_order10_pack_auditor.py \
  --manifest positive-square-energy/experiments/rank6_order10_search_manifest.json \
  --chunk-index i --write-chunk-transcript /tmp/r10-replay/chunk-i.json
```

The literal `i` is replaced by the decimal index. The order-ten command also launches and compares optimized replay; order nine must be run explicitly under both `python3` and `python3 -O` until its auditor provides the same built-in comparison. Each segment authenticates the full manifest and

dependency map before auditing its own raw records, so segment boundaries cannot alter certificate semantics. A clean full replay requires all chunks, then the full auditor with no `--chunk-index`; if coverage is incomplete or any target remains unowned, the expected result is nonzero status. Receipts and transcripts are resumability aids only and are not theorem owners.

B.7 Hash and acceptance placeholders

The frozen kernel-fixture SHA-256 currently embedded in both streams is `5a862a0e9ed5dfe91ff6f8491936c8e775`. All other promotion identities below deliberately remain placeholders; they must be filled from a clean committed replay and checked by a master, not copied from an experimental run.

identity	required value before promotion
R9G1 generator/auditor sources	TODO-R9-SOURCE-SHA256-MAP
R9G1 manifest and exact report	TODO-R9-MANIFEST-SHA256, TODO-R9-REPORT-SHA256
R10G1 generator/auditor sources	TODO-R10-SOURCE-SHA256-MAP
R10G1 manifest and exact report	TODO-R10-MANIFEST-SHA256, TODO-R10-REPORT-SHA256
symbolic fixtures and recognizers	TODO-SYMBOLIC-SOURCE-AND-FIXTURE-SHA256-MAP
orders-2–10 master and transitive root	TODO-RANK6-MASTER-SOURCE-SHA256, TODO-RANK6-TRANSITIVE-ROOT
all-connected synthesis master	TODO-ALL-CONNECTED-SOURCE-SHA256, TODO-ALL-CONNECTED-REPORT-SHA256

Each source map must name every relative path and hash its exact bytes; a single digest with no leaf listing is insufficient. Chunk compressed/raw hashes and ranges live in the canonical manifests and must be reproduced there rather than typeset as a mutable table. No placeholder, hash, receipt, or present partial manifest discharges Premise 2 of Theorem 1.1.

C Trusted base and limitations

The analytic trust base consists of the displayed matrix arguments, exact path elimination, standard block theory, the suppression reduction, and every structural lemma imported through Premise 1 or a typed finite owner. Those imported repository notes are mathematical dependencies, not external validation. The computer-assisted trust base additionally includes the committed verifier and generator sources, canonical kernel fixtures, every certificate and chunk consumed transitively, decompression and parsing code, exact integer/rational and validated symbolic semantics, canonical labelling, the Python runtime, and executing hardware. Hashes establish byte identity, not truth; normal/optimized agreement detects only some implementation errors; hostile mutations are tests, not a completeness proof; and no independent implementation is claimed. Numerical optimization may discover witnesses but is not a proof object. Rooted-tree attachment means a genuine one-vertex sum; a connector meeting a core twice is outside that lemma. Kernel parallel edges are permitted only before realization; the graph to which the trace identity is applied remains simple.

AI disclosure and nonpublication notice

This draft was generated by an AI coding and mathematical reasoning system from the repository materials cited below. No human authorship, human review, independent verification, peer review,

acceptance, publication, or theorem announcement is claimed. The manuscript is a conditional internal skeleton pending the explicit promotion gates in Appendix A.

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